

Solution Code - Module 3-Core Java - TypeCastingExample

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q07_TypeCastingExample.java

```
1 | public class Q07_TypeCastingExample {
2 |     public static void main(String[] args) {
3 |         // Double to int (Narrowing/Explicit casting)
4 |         double originalDouble = 99.99;
5 |         int castedInt = (int) originalDouble;
6 |         System.out.println("Original double value: " + originalDouble);
7 |         System.out.println("Casted int value (narrowing): " + castedInt);
8 |
9 |         // Int to double (Widening/Implicit casting)
10 |        int originalInt = 42;
11 |        double castedDouble = originalInt; // implicit casting
12 |        System.out.println("\nOriginal int value: " + originalInt);
13 |        System.out.println("Casted double value (widening): " + castedDouble);
14 |
15 |        // Explicitly casting int to double for demonstration
16 |        double explicitCastedDouble = (double) originalInt;
17 |        System.out.println("Explicitly casted double value: " + explicitCastedDouble);
18 |    }
19 | }
```